

减压稳压阀1803D2系列 Gas regulator valve 1803D2 series

用心铸配·中国制造
Made in China by Heart

1803D2系列减压阀广泛应用于一级减压系统,为市场应用最多的高中压减压阀

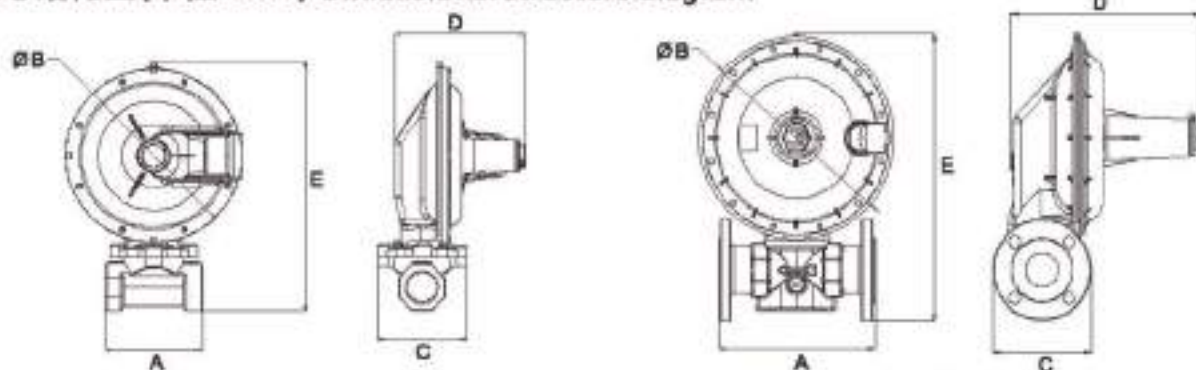
1803D2 series pressure reducing valve is widely used in the level of the relief system, the most high pressure reducing valve for the market application

技术参数 technical parameter

- ▶ 口径: DN20、DN25、DN32、DN40、DN50
- ▶ 最大入口压力: 10bar
- ▶ 最大流量达: 700Nm³/h
- ▶ 灵敏的关闭性
- ▶ 调节精度高
- ▶ 可安装于任何位置
- ▶ 可作大型调压器之负载调压
- ▶ 压力设定简单, 不同弹簧满足不同的压力范围
- ▶ 适用于天然气、液化石油气、空气、氮气和它气体
- ▶ CAL: DN20、DN25、DN32、DN40、DN50
- ▶ Biggest inlet pressure: 10 bar
- ▶ Maximum flow: 700 Nm³/h
- ▶ Sensitive to shut down
- ▶ Adjust the high precision
- ▶ Can be installed in any position
- ▶ As a large load voltage regulation of voltage regulator
- ▶ Simple pressure setting, different spring meet different pressure range
- ▶ Suitable for natural gas, liquefied petroleum gas, water, dry air, nitrogen and other gases



尺寸结构图(单位:mm) Dimensional structure diagram



型号 Model	口径 CAL	连接方式 Connection mode	最大进口压力(bar) Maximum import pressure	A	B	C	D	E
1803D2DN20	20	Rp3/4"	10	102	184	93	144	265
1803D2DN25	25	Rp1"	10	102	184	93	144	265
1803D2DN32	32	Rp1 1/4"	10	102	184	93	144	268
1803D2DN40R	40	Rp1 1/2"	10	118	184	93	144	275
1803D2DN50R	50	Rp2"	10	118	184	93	144	287
1803D2DN40	40	Rp1 1/2"	10	152	318	85	310	435
1803D2DN50	50	Rp2"	10	152	318	85	310	435
1803D2DN40F	40	法兰连接	10	255	318	160	310	450
1803D2DN50F	50	法兰连接	10	255	184	160	310	450



1803D2 and 1803D2-HC Series Service Regulators

General Information

The 1803D2 and 1803D2-HC regulators are available as full capacity internal relief models. Series 1803D2 and 1803D2-HC are also available with over and underpressure shut-off protection.

Available valve body sizes: 3/4" x 3/4", 3/4" x 1", 3/4" x 1-1/4", 1" x 1", 1" x 1-1/4", 1-1/4" x 1-1/4" NPT or BSP-TR.

Also available as offset valve body: 3/4" x 3/4", 3/4" x 1", 1" x 1" NPT or BSP-TR.

1803D2-HC is available with 1-1/4" valve body only.

Maximum inlet pressure, normal service (P1):
Depending on orifice size, up to 125 PSIG.

Maximum outlet set pressure, normal service (P2):
Depending on spring selection, up to 2 PSIG (56" W.C.).

Orifice Sizes

Orifice Size*	Part Number	
	Standard	w/OPSO
9/16"	72494P026	72751P019
1/2"	72494P025	72751P016
3/8"	72494P023	72751P014
5/16"	72494P022	72751P013
UPSO	71422G004	71422G004
1/4"	72494P021	72751P012
3/16"	72494P020	72751P011
1/8" x 3/16"	72494P030	72751P020
1/8"	72494P019	N/A

* See SB 8510 for maximum inlet recommendations and capacity performance for each orifice size.

Pressure Springs

Outlet Pressure Range	Color Code	Part Number
3.5" to 6" W.C.	Blue	70017P043
3.5" to 7.5" W.C.	Tan	70017P089
5.5" to 8.5" W.C.	Yellow	70017P044
6" to 12" W.C.	Brown	70017P137
6" to 15" W.C.	Purple	70017P042
12" to 28" W.C.	White	70017P080
24" to 48" W.C.	Red	70017P082
42" W.C. to 2.0 PSIG	Red-Red	70017P049

Preparations

CAUTION

Do not connect the inlet of the regulator to a normal pressure source higher than that recommended. Never connect the regulator outlet to the source of pressure. Observe the flow direction arrow on the valve body. If inlet pressure can exceed the maximum regulator outlet pressure rating, some form of over-pressure protection in addition to internal relief may be required. To prevent damage to the regulator and possible personal injury, inlet and outlet pressures must not exceed the regulator pressure rating stated on page 4.

Install, operate, inspect and maintain the regulator as outlined in the following instructions and in accordance with your company's policies and applicable federal, state and local codes and laws. Failure to follow these instructions may result in damage to the regulator or personal injury.

1. Check installation location for suitability. It is possible that physical changes have been made to the building site since the regulator and location were originally selected.
2. Examine regulator for shipping damage.
3. Check regulator and piping for foreign matter which may have accumulated during shipment or handling.
4. Check seal plug information to determine if regulator is suitable for intended service.
5. Suitable stop valve(s) should be installed and should be conveniently located.
6. Filters are available from American Meter for applications where pipeline contaminants are suspected to be present.
7. Regulators may be used at temperatures between -20° F and +150° F (-30° C and +65° C).

Installation

(Numbers in parentheses refer to Figure 1, page 3.)

WARNING

Inside Installation:

Will require a vent line of sufficient diameter to carry gas vented by the regulator to a safe outside location away from any opening in the building. For all sizes of regulators, a 1" NPT or BSP-TR is provided for a vent line. Do not thread a pipe into the vent more than seven turns so as not to block the vent flapper.

Outside Installation:

Will require that care be taken to prevent vent opening from freezing closed or becoming blocked or permitting water to enter from any cause. Particular consideration should be given to sites where flooding, snow, or freezing rain may be experienced. The vent or vent-line port should point vertically downward, and overhead protection should be used where necessary.

1. Remove all shipping plugs.
2. Use good piping practice. Be sure piping and regulators are free of dirt, pipe dope and other debris. Apply pipe dope to male threads only.
3. Install regulator. Make certain it is piped correctly with inlet pipe connected to the inlet regulator connection and flow is in the direction as indicated by arrow located on valve body (19). The vent should be positioned to prevent entry of water and debris. (American Meter recommends that the vent face downward.) By removing the two bolts (5), the diaphragm case (8) may be rotated in 90-degree increments in relation to the valve body (19). Replace and tighten the bolts (5) to 100 ±25 in. lbs. For inside installations, a regulator installed within a building should be located as near as practical to the point of service line entrance.
4. Turn gas on slowly. If an outlet stop valve is used, it should be open first. Do not overload the diaphragm with a sudden surge of inlet pressure.

1803D2 and 1803D2-HC Series Service Regulators

Installation *continued*

5. Assure that there are no leaks and all connections are tight.
6. The 1803D2 and 1803D2-HC Series regulator is pre-set at the factory with the diaphragm in the horizontal position and should not require any adjustment if installed in that position. The outlet pressure may vary as much as 0.5" W.C. if the regulator is installed in the vertical position. Should you decide to adjust the set pressure, perform the following:
 - a. Turn gas off and depressurize the system.
 - b. Install outlet pressure gauge. Remove seal plug (14).
 - c. Turn gas on slowly.
 - d. Establish a low flow rate - 45 to 55 SCFH.
 - e. Turn pressure spring adjustment screw (9) clockwise to increase outlet pressure or counterclockwise to decrease outlet pressure.
 - f. Establish the set pressure within the range limits of the pressure spring used.
 - g. To test for the regulator's ability to fully shut off (lock-up), shut off all flow downstream of the regulator. At lock-up, the outlet pressure will be somewhat higher than the set pressure. However, if the pressure continues to rise after 3 seconds, the regulator must be repaired. Observe for 30 seconds.
 - h. When the set pressure has been properly adjusted, depressurize and remove gauge(s) and lines.
 - i. Replace seal plug (14).
 - j. Pressurize and check all connections for leaks.

Inspection

(Numbers in parentheses refer to Figure 1.)

Inspection and maintenance of the 1803D2 and 1803D2-HC Series regulator have been simplified for practical reasons. Repair parts are offered if maintenance of the regulator is required. Inspection of the valve seat disc (1) and orifice valve (7) may be performed in the field. Follow the procedure described below. Care should be exercised to prevent foreign matter from entering the regulator.

Inspection of Valve Seat Disc and Orifice

1. Shut off gas supply and depressurize the regulator system.
2. Remove two bolts (5), securing the diaphragm case (8) to the valve body (19). Separate the diaphragm case from the valve body.
3. Visually check the seating edge of the orifice (7). If it is nicked or damaged, it should be replaced with a new orifice to provide proper shut off. See Maintenance instructions for replacing orifice.
4. Inspect the surface of the valve seat disc (1). If it is scored or uneven, replace it as outlined in the Maintenance instructions.
5. Install new seal ring (2) in the diaphragm case groove.
6. Inspect unit for cleanliness and proper positioning of the parts. Pay particular attention to the proper positioning of the seal ring (2) in the groove.

7. Install the diaphragm case (8) on the valve body (19). Make sure the vent is pointed downward. Replace and tighten the valve body bolts (5) to 100 $\pm$ 25 in. lbs.
8. Proceed to Steps 4, 5 and 6 of the Installation instructions.

CAUTION

Regulators are pressure-control mechanisms having numerous moving parts which can wear. In addition, regulator damage may occur from external sources. For these reasons, the regulator should be periodically inspected and checked for proper operation. The frequency of inspection will depend on the severity of the service conditions and the requirements of applicable local, state and federal codes and regulations.

Maintenance

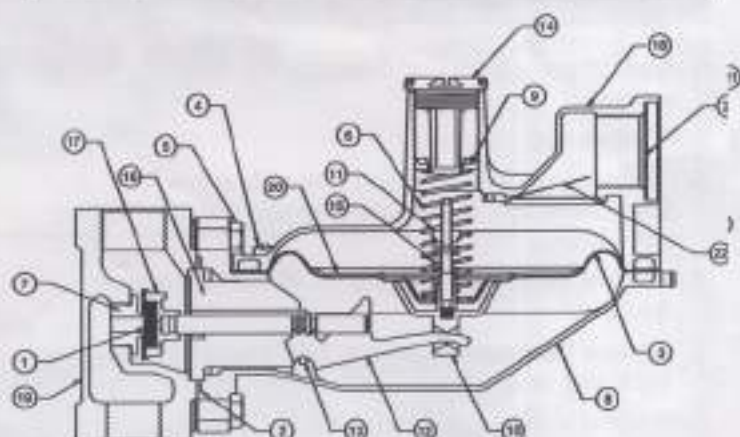
(Numbers in parentheses refer to Figure 1 below.)

Replacing Seat Disc

1. Carefully insert the tip of a knife along the edge of the seat disc (1) and lift the seat disc out of the plunger (17).
2. Install a new seat disc (1) into the plunger (17) by pressing into place. Make sure that the disc is pressed completely into the plunger (17).
3. Proceed to Steps 5, 6, 7 and 8 of the Inspection instructions.

CAUTION - Handling and Disposal

"As a knowledgeable user of American Meter Company products, you should be aware that parts in the Company's meters and regulators contain or are coated with heavy metals such as cadmium, zinc, lead and chromium. Obviously, therefore, repair or refurbishment of this equipment should take into account the presence of these materials and should comply with all state and federal requirements concerning worker protection, proper disposal and safety, including protection against exposure to dust and fumes."



Item No.	Description
1	Disc, Valve Seat
2	Seal Ring, Valve Body
3	Diaphragm
4	Screw, Diaphragm Case
5	Bolt, Valve Body
6	Spring, Pressure
7	Valve, Orifice
8	Diaphragm Case
9	Screw, Pressure Adj.
10	Stem
11	Nut, Relief Valve

Item No.	Description
12	Lever
13	Pin, Lever
14	Plug, Seal
15	Spring, Relief Valve
16	Guide, Valve Plunger
17	Plunger, Valve
18	Assembly, Top
19	Valve, Body
20	Plate, Diaphragm
21	Screen, Vent
22	Vent Valve